J Nutr Health Aging
Volume 19, Number 7, 2015

THE RAPID COGNITIVE SCREEN (RCS): A POINT-OF-CARE SCREENING FOR DEMENTIA AND MILD COGNITIVE IMPAIRMENT

T.K. MALMSTROM[1,2], V.B. VOSS[3], D.M. CRUZ-OLIVER[2], L.A. CUMMINGS-VAUGHN[4,5], N. TUMOSA[4], G.T. GROSSBERG[1], J.E. MORLEY[2]

1. Department of Neurology & Psychiatry, Saint Louis University School of Medicine, St. Louis, Missouri, USA; 2. Division of Geriatric Medicine, Saint Louis University School of Medicine, St. Louis, Missouri, USA; 3. Saint Louis University School of Medicine, St. Louis, Missouri, USA; 4. Jefferson Barracks Division, Geriatric Research Education and Clinical Center, Veterans Affairs Saint Louis Health System, St. Louis, Missouri, USA; 5. Division of Geriatrics and Nutritional Science, Washington University in St. Louis, St. Louis, Missouri, USA. Corresponding author: Theodore K. Malmstrom, PhD, Department of Neurology & Psychiatry, School of Medicine, Saint Louis University, 1438 South Grand Boulevard, St. Louis, Missouri 63104, Email: malmsttk@slu.edu; Phone: 314-977-4814; Fax: 314-977-4879

Abstract: *Objectives:* There is a need for a rapid screening test for mild cognitive impairment (MCI) and dementia to be used by primary care physicians. The Rapid Cognitive Screen (RCS) is a brief screening tool (< 3 min) for cognitive dysfunction. RCS includes 3-items from the Veterans Affairs Saint Louis University Mental Status (SLUMS) exam: recall, clock drawing, and insight. Study objectives were to: 1) examine the RCS sensitivity and specificity for MCI and dementia, 2) evaluate the RCS predictive validity for nursing home placement and mortality, and 3) compare the RCS to the clock drawing test (CDT) plus recall. *Methods:* Patients were recruited from the St. Louis, MO Geriatric Research Education and Clinical Center (GRECC), Veterans Affairs Medical Center (VAMC) hospitals (study 1) or the Saint Louis University Geriatric Medicine and Psychiatry outpatient clinics (study 2). Study 1 participants (N=702; ages 65-92) completed cognitive evaluations and 76% (n=533/706) were followed up to 7.5 years for nursing home placement and mortality. Receiver operator characteristic (ROC) curves were computed to determine sensitivity and specificity for MCI (n=180) and dementia (n=82). Logistic regressions were computed for nursing home placement (n=31) and mortality (n=176). Study 2 participants (N=168; ages 60-90) completed the RCS and SLUMS exam. ROC curves were computed to determine sensitivity and specificity for MCI (n=61) and dementia (n=74). *Results:* RCS predicted dementia and MCI in study 1 with optimal cutoff scores of ≤ 5 for dementia (sensitivity=0.89, specificity=0.94) and ≤ 7 for MCI (sensitivity=0.87, specificity=0.70). The CDT plus recall predicted dementia and MCI in study 1 with optimal cutoff scores of ≤ 2 for dementia (sensitivity=0.87, specificity=0.85) and ≤ 3 for MCI (sensitivity=0.62, specificity=0.62). Higher RCS scores were protective against nursing home placement and mortality. The RCS predicted dementia and MCI in study 2. *Conclusions:* The 3-item RCS exhibits good sensitivity and specificity for the detection of MCI and dementia, and higher cognitive function on the RCS is protective against nursing home placement and mortality. The RCS may be a useful screening instrument for the detection of cognitive dysfunction in the primary care setting.

Key words: Screening, dementia, mild cognitive impairment, Alzheimer's disease, cognition.

One in nine persons aged 65 years or older and 32% of persons over 85 years are estimated to have Alzheimer's disease (1). By 2050 the number of persons with Alzheimer's disease in the United States is predicted to be 13.8 million. Primary care physicians frequently do not identify persons with cognitive dysfunction (2-5). Ideally a screening tool for primary care should be easy to administer, brief (less than 5 minutes), valid, free to use, and exhibit low bias due to education, language, and culture (6). There is presently no gold standard screening tool for the detection of cognitive dysfunction (6).

Mild cognitive impairment (MCI) is characterized by cognitive impairment with intact functional abilities; it represents an intermediate condition between normal cognitive impairment of aging and persons with dementia (7-8). There are a number of treatable conditions that produce MCI (9). There are several diagnostic guidelines for mild cognitive impairment (MCI), including the National Institute on Aging-Alzheimer's Association (NIA-AA) (10), Diagnostic and Statistical Manual of Mental Disorders (5th edition; DSM-V) (11), and the International Work Group (IWG) (12) criteria. Depending on the definition used, there are wide variations in prevalence of MCI, ranging from 3.6% to 38.4% in persons over 70 years of age (13). MCI is recognized as a risk factor for Alzheimer's disease, though the majority of persons with MCI either do not progress and some exhibit improved cognition (8).

A number of screening tests for dementia and MCI exist (14). Many MCI screening instruments take over 5 minutes to administer, which may be an excessive amount of time for a busy primary care physician. A few, such as the clock drawing test (CDT) and Mini-Cog (15, 16), take less than 5 minutes to administer, which is a more reasonable time period for primary care physicians to administer a screening test. However, many very brief screening measures do not have good sensitivity and specificity for the diagnosis of MCI (17).

The Saint Louis University Mental Status (SLUMS) exam is a screening test with good sensitivity and specificity for both dementia and MCI (18-22). The SLUMS exam has been translated into many languages (available at http://aging.slu.edu/). It takes approximately 6 to 8 minutes to administer. In this paper we investigate the performance of an abbreviated

Received June 23, 2015
Accepted for publication June 26, 2015

form of the SLUMS exam, the Rapid Cognitive Screen (RCS) that takes < 3 minutes to administer. Study objectives are to: 1) examine the RCS sensitivity and specificity to detect MCI and dementia on the DSM and on the SLUMS exam, 2) evaluate the RCS predictive validity for nursing home placement and mortality, and 3) compare the sensitivity and specificity of the RCS to the CDT plus recall.

Methods

The RCS includes 3-items from the Veterans Affairs Saint Louis University Mental Status (SLUMS) exam: recall of 5 words, a clock drawing test, and the ability to remember a story and convert the fact that Chicago is in Illinois (Figure 1). RCS scores range from 0=worst to 10=best, including memory (5 points), clock drawing (4 points; 2 points for hour markers, 2 points for time), and story (1 point). The CDT plus recall consists of clock drawing (2 points) and recall of 3 words (3 points).

Figure 1
Rapid Cognitive Screen (RCS)

Name_____ Age_____

Is the patient alert?_____ Level of education_____

1. **Please remember these five objects. I will ask you what they are later.**
[Read each object to patient using approximately 1 second intervals.]

Apple **Pen** **Tie** **House** **Car**

Please repeat the objects for me. [If patient does not repeat all 5 objects correctly, repeat until all objects are recalled correctly or up to a maximum of 2 times.]

2. [Give patient pencil and the blank sheet with clock face.]
This is a clock face. Please put in the hour markers and the time at ten minutes to eleven o'clock.

____/2 (points) Hour markers okay
____/2 (points) Time correct

[When scoring, give full credit for all 12 numbers. If the patient puts only ticks on the circle, prompt them once to put numbers next to those ticks for full credit. Do not repeat the time. When scoring the correct time, make sure that the minute hand points at the 10 and the hour hand points at the 11.]

3. **What were the five objects I asked you to remember?**

____/1 (point) Apple
____/1 (point) Pen
____/1 (point) Tie
____/1 (point) House
____/1 (point) Car

4. **I'm going to tell you a story. Please listen carefully because afterwards, I'm going to ask you about it.**

Jill was a very successful stockbroker. She made a lot of money on the stock market. She then met Jack, a devastatingly handsome man. She married him and had three children. They lived in Chicago. She then stopped work and stayed at home to bring up her children. When they were teenagers, she went back to work. She and Jack lived happily ever after.

What state did she live in?

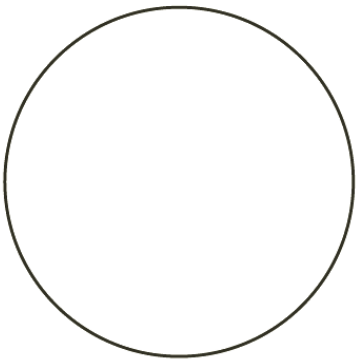
____/1 (point) Illinois

[Do not repeat the story but do make sure the patient is paying attention the first time you read it to them. Do not prompt or give hints. The answer of "Chicago" as the state she lives in gets no credit but you may prompt them once by repeating the question when "Chicago" is given as the answer.]

_____ Total Score [0-10 points]

SCORING
8-10…….. Normal
6-7……. Mild Cognitive Impairment
0-5……. Dementia

_____ _____ _____
CLINICIAN'S SIGNATURE DATE TIME



In our first study the ability of the RCS to discriminate between normal cognition, MCI, and dementia on the DSM-IV; to predict nursing home placement and mortality; and to perform as well or better than the CDT plus recall were evaluated using our database (18). DSM-IV criteria were used to make the diagnosis of mild neurocognitive disorder or dementia. The sample included N=702 participants ages 65-92 from the Saint Louis Veterans Affairs Medical Center hospitals recruited in 2003. Dementia was present in 12% (n=82) of participants and 26% (n=180) participants had MCI. All participants were males (100%). Participant follow-up to evaluate the association of RCS scores with nursing home placement and mortality up to 7.5 years after completing the RCS included N=533/702 participants. Participants (n=169/702) were lost to follow-up due to a change in primary care provider, incorrect contact information or inactive electronic medical record. There were no differences in age, gender, or education for participants lost to follow-up, compared to those with valid follow-up data for nursing home placement and mortality.

In a second study we examined the ability of the RCS to detect dementia and MCI in an outpatient setting. The sample included N=168 participants ages 60-90 from the Saint Louis University Geriatric Medicine and Geriatric Psychiatry outpatient clinics in 2013. There were 44% (n=74) participants with dementia and 36% (n=61) participants with MCI. The sample was 62% (n=104) female. African Americans were 42% (n=71) of the participants and Caucasians were 55% (n=92).

Statistics

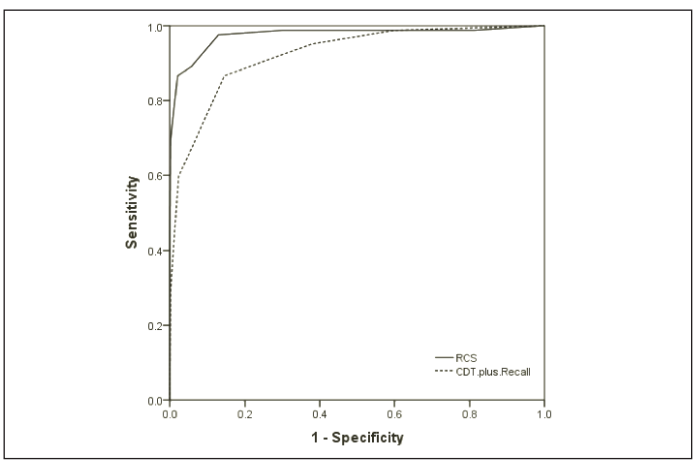
Data were analyzed using IBM SPSS Statistics, version 22 (Somers, NY) and SAS, version 9.2 (SAS Institute, Inc., Cary, NC) for ROC contrasts. Descriptive statistics are reported as means ± standard deviations (SD) or percentages. Scale total scores for the RCS (0-10) and CDT plus recall (0-5) were used for ROC curves for MCI and dementia. Using a standard approach, sensitivity and specificity were calculated for the RCS and CDT plus recall. ROC contrasts were computed to compare the total area under the curve (AUC) of the screening tests (RCS vs. CDT plus recall) for MCI and for dementia on the DSM-IV. Logistic regression was used to investigate the

associations of the RCS with nursing home placement and mortality. Odds ratios (ORs) and 95% confidence intervals (CIs) adjusted for age are reported for logistic regression analyses.

Results

In study one, the RCS predicted dementia (AUC=0.98, 95% CI: 0.95-1.00) and MCI (AUC=0.86, 95% CI: 0.83-0.89) on the DSM-IV. RCS optimal cutoff scores were ≤ 5 for dementia (sensitivity=0.89, specificity=0.94) and ≤ 7 for MCI (sensitivity=0.87, specificity=0.70). The CDT plus recall also predicted dementia (AUC=0.92, 95% CI: 0.89-0.96) and MCI (AUC=0.66, 95% CI: 0.61-0.70) on the DSM-IV. CDT plus recall optimal cutoff scores were ≤ 2 for dementia (sensitivity=0.87, specificity=0.85) and ≤ 3 for MCI (sensitivity=0.62, specificity=0.62). The RCS and CDT plus recall ROC curves for dementia based on the DSM-IV are displayed in Figure 2a. A ROC contrast comparing the AUCs for dementia shows that the RCS (AUC=0.98) performed better than the CDT plus recall (AUC=0.92) ($p < .01$). The RCS and CDT plus recall ROC curves for MCI on the DSM-IV are displayed in Figure 2b. A ROC contrast comparing the AUCs for MCI shows that the RCS (AUC=0.86) was superior to the CDT plus recall (AUC=0.66) ($p<.001$). Higher RCS scores (0-10) were protective against nursing home placement (OR per point on the RCS=0.85, 95% CI: 0.73-0.98) and mortality (OR per point on the RCS=0.84, 95% CI: 0.77-0.91) up to 7.5 years later.

Figure 2a
RCS & CDT Plus Recall ROC Curves for Dementia on the DSM-IV



In study two, the RCS also predicted dementia (AUC=0.97, 95% CI: 0.94-0.99) and MCI (AUC=0.79, 95% CI: 0.69-0.88) on the SLUMS exam. Figures 2c and 2d display the RCS ROC curves for dementia and MCI, respectively, on the SLUMS exam. RCS optimal cutoff scores were ≤ 5 for dementia (sensitivity=0.92, specificity=0.94) on the SLUMS exam and ≤ 7 for MCI (sensitivity=0.69, specificity=0.82).

Figure 2b
RCS and CDT Plus Recall ROC Curves for MCI on the DSM-IV

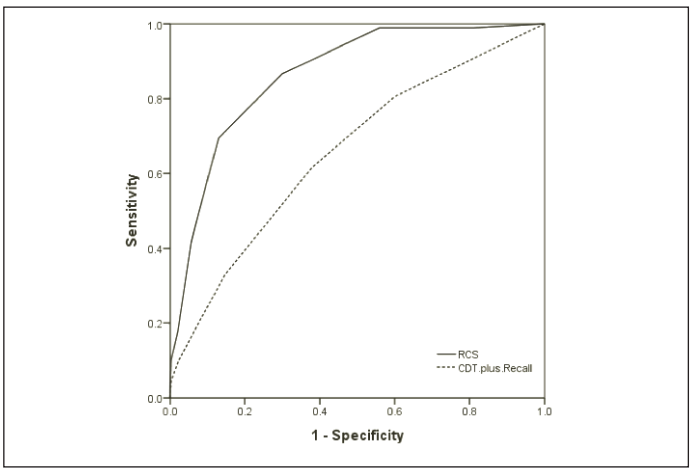
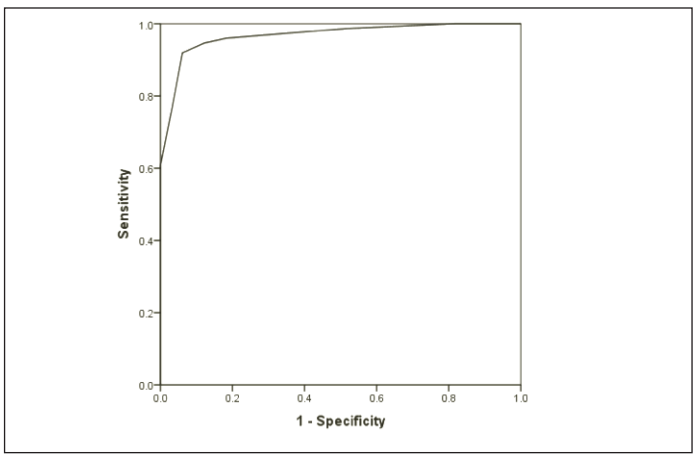


Figure 2c
RCS ROC Curve for Dementia on the SLUMS Exam

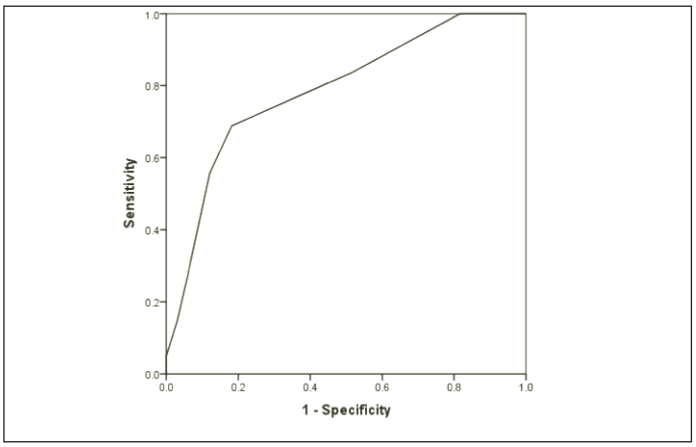


Discussion

Our objective was to assess the performance of a much needed brief screening tool for cognitive impairment. The RCS exhibited construct validity to screen for dementia and MCI in both studies. Study one showed that the 3-item RCS and the 2-item CDT plus recall have high sensitivity and specificity for detecting dementia on the DSM-IV, and study two showed that the RCS also has very good sensitivity and specificity for the detection of dementia on the SLUMS exam. The optimal scores for dementia on the RCS were ≤ 5 in both studies, and on the CDT plus recall were ≤ 2. The RCS exhibited good sensitive and specificity for detecting MCI on the DSM-IV (study one) and SLUMS (study two) with optimal scores of ≤ 7. The CDT plus recall exhibited only modest sensitivity and specificity for the detection of MCI on the DSM-IV with optimal scores of ≤ 3. ROC contrasts for the AUC showed that the RCS is a better predictor overall of dementia and MCI on the DSM-IV with

the largest difference observed for MCI. The RCS predictive validity for institutionalization and mortality after 7.5 years was also investigated in study one. Persons without RCS scores in the MCI and/or dementia range were less likely to be institutionalized or die over 7.5 years.

Figure 2d
RCS ROC Curve for MCI on the SLUMS Exam



The RCS and the CDT plus recall are both very brief screening instruments that can be administered in < 3 minutes in a busy clinical setting. The CDT alone has previously been shown to be a good measure to screening for dementia (23). This study provides preliminary evidence that the RCS may also be an effective measure to screen for dementia. The RCS may be a better tool than the CDT plus recall for detecting MCI based on the results of our study; the CDT alone has not been studied as a measure to detect MCI. Previous studies have shown mixed results regarding the utility of the Mini-Cog to detect MCI (24, 25). More evidence and further study of the RCS as a screening tool for cognitive impairment are warranted based on the studies reported herein.

The RCS is valid, very brief (< 3 minutes), easy to administer and score, and has no copyright. This study provides preliminary evidence that the RCS may be a useful screening instrument for the detection of cognitive impairment in a busy clinical setting.

Conflicts of Interest: The authors have nothing to disclose.

Ethical standard: This research was approved by the Institutional Review Boards of the St. Louis VAMC (study 1) and Saint Louis University (study 2).

References

1. Alzheimer's Association. Alzheimer's Association Report: 2015 Alzheimer's disease facts and figures. Alzheimers Dement 2015;11:332-384.
2. Chodish J, Petitti DB, Elliott M, et al. Physician recognition of cognitive impairment: Evaluating the need for improvement. J Am Geriatr Soc 2004;52:1051-1059.
3. Callahan CM, Hendrie HC, Tierney WM. Documentation and evaluation of cognitive impairment in elderly primary care patients. Ann Inter Med 1995;122:422-429.
4. Valcour FB, Masaki KH, Curb JD, Blanchette PL. The detection of dementia in the primary care setting. Arch Intern Med 2000;160:2964-2968.
5. Boustani M, Callahan CM, Unverzagt FW, et al. Implementing a screening and diagnosis is program for dementia in primary care. J Gen Intern Med 2005;20:572-577.
6. Cordell CC, Borson S, Boustani M, et al. Alzheimer's Association recommendations for operationalizing the detection of cognitive impairment during the Medicare Annual Wellness Visit in a primary care setting. Alzheimers Dement 2013;9:141-150.
7. Peterson RC. Mild cognitive impairment. N Engl J Med 2011;364;23:2227-2234.
8. Langa, KM, Levine DA. The diagnosis and management of mild cognitive impairment. JAMA 2014;312:2251-2561.
9. Morley JE. Mild cognitive impairment – a treatable condition. J Am Med Dir Assoc 2014;15:1-5.
10. Albert MS, DeKosky ST, Dickson D, et al. The diagnosis of mild cognitive impairment due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guideline for Alzheimer's disease. Alzheimers Dement 2011;7:270-279.
11. American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 5th edition. Arlington, VA: American Psychiatric Publishing; 2013.
12. Dubois B, Feldman HH, Jacova C, et al. Revising the definition of Alzheimer's disease: A new lexicon. Lancet Neurol 2010;9:1118-1127.
13. Ward A, Arrighi HM, Michels S, Cedarbaum JM. Mild cognitive impairment: Disparity of incidence and prevalence estimates. Alzheimers Dement. 2012;8:14-21.
14. Lin JS, O'Connor E, Rossom R, et al. Screening for cognitive impairment in older adults: An evidence update for the U.S. Preventive Services Task Force. Rockville, MD: Agency for Healthcare Research and Quality; 2013. Evidence Report 107. AHRQ Publication 14-05198-EF-1.
15. McCarten JR, Anderson P, Kuskowski MA, et al. Screening for cognitive impairment in an elderly veteran population: Acceptability and results using different versions of the Mini-Cog. J Am Geriatr Soc 2011;59:309-313.
16. Borson S, Scanlan JM, Chen P, Ganguli M. The Mini-Cog as a screen for dementia: Validation in a population-based sample. J Am Geriatr Soc 2003;51:1451-1454.
17. Lonie JA, Tierney KM, Ebmeier KP. Screening for mild cognitive impairment: A systematic review. Int J Geriatr Psychiatry 2009;24:902-915.
18. Tariq SH, Tumosa N, Chibnall JT, et al. Comparison of the Saint Louis University Mental Status Examination and the Mini-Mental State Examination for detecting dementia and mild neurocognitive disorder—a pilot study. Am J Geriatr Psychiatry 2006;14:900-910.
19. Cummings-Vaughn LA, Chavakula NN, Malmstrom TK, et al. Veterans Affairs Saint Louis University Mental Status examination compared with the Montreal Cognitive Assessment and the Short Test of Mental Status. J Am Geriatr Soc 2014;62:1341-1346.
20. Feliciano L, Horning SM, Klebe KJ, et al. Utility of the SLUMS as a cognitive screening tool among a nonveteran sample of older adults. Am J Geriatr Psychiatry 2013;21:623-630.
21. Cao L, Hai S, Lin X, et al. Comparison of the Saint Louis University Mental Status Examination, the Mini-Mental Status Examination, and the Montreal Cognitive Assessment in detection of cognitive impairment in Chinese elderly from the geriatric department. J Am Med Dir Assoc 2012;13:626-629.
22. Cruz-Oliver DM, Malmstrom TK, Allen CM, et al. The Veterans Affairs Saint Louis University Mental Status exam (SLUMS exam) and the Mini-mental status exam as predictors of mortality and institutionalization. J Nutr Health Aging 2012;16:636-641.
23. Kaufer DI, Williams CS, Braaten AJ, et al. Cognitive screening for dementia and mild cognitive impairment in assisted living: Comparison of 3 tests. J Am Med Dir Assoc 2008;9:586-593.
24. Holsinger T, Plassman BL, Stechuchak KM, et al. Screening for cognitive impairment: Comparing the performance of four instruments in primary care. J Am Geriatr Soc 2012;60:1027-1036.
25. Borson S, Scanlan JM, Watanabe J, et al. Improving identification of cognitive impairment in primary care. Int J Geriatr Psychiatry 2006;21:349-355.